



WATER JUG PROBLEM 5

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You can perform the following operations any number of times:

Fill either jug completely. Empty either jug completely. Pour water from one jug into the other until either the source jug is empty or the destination jug is full.

Given capacities A, B, and a target amount T, determine whether it is possible to measure exactly T liters of water in either jug.

Input Format

The first line contains an integer Q, the number of test cases.

Each of the next Q lines contains three space-separated integers:

A B T

A — capacity of the first jug. B — capacity of the second jug. T — required amount of water.

Constraints

$1 \leq Q \leq 100$ $1 \leq A, B \leq 10^9$ $0 \leq T \leq 10^9$

Output Format

For each test case, print:

YES

if exactly T liters can be measured, otherwise print:

NO

Sample Input 0

```
5 3 4
```

Sample Output 0

```
YES
```

Sample Input 1

```
5 10 3
```

Sample Output 1

```
NO
```



```
1 import math
2
3 A, B, T = map(int, input().split())
4
5 if T == 0:
6     print("YES")
7 elif T <= max(A, B) and T % math.gcd(A, B) == 0:
8     print("YES")
9 else:
10     print("NO")
11
```

Line: 11 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

 **Sample Test case 0**[Download](#) **Sample Test case 1**

Input (stdin)

1	5	3	4
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Your Output (stdout)

1	YES
---	-----

Expected Output

1	YES
---	-----

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